IX: A Tractable Descendant of AIXI for Autonomous Capital Compounding

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***Abstract*—We introduce IX, a tractable architecture for an autonomous agent whose objective is recursive self-financing (RSF): the systematic deployment of recurring operator capital contributions and the reinvestment of agent-generated returns toward a specified terminal liquidity target. IX inherits its theoretical scaffolding from Hutter's AIXI and its computational pattern from the MC-AIXI-CTW approximation of Veness *et al.*, substituting AIXI's uncomputable universal prior with a domain-specific Bayesian world model over market microstructure, and substituting its expectimax planner with policy-guided Monte Carlo tree search under an explicit ruin-probability constraint. We propose IX-Bench v0.1, an eight-family benchmark suite that grades agents on returns, risk, recursion quality, and probability-of-target-achievement under stated capital constraints. We outline a seven-phase implementation pipeline mapped to USPTO prior-art clusters and open-source repository anchors. We close with a non-optional treatment of empirical reality: a 2.5% monthly compound return (34.5% annualized) sits in the top decile of documented systematic returns—below Renaissance Medallion's ∼40% lifetime average but materially above commodity quant benchmarks. The benchmark is structured to surface the gap between assumed and achieved rather than obscure it.**

***Index Terms*—AIXI, MC-AIXI-CTW, recursive self-financing, autonomous agents, Monte Carlo tree search, Kelly criterion, algorithmic trading, USPTO prior art**

I. Introduction

A recursively self-financing agent is a closed-loop system that takes positions in markets, reinvests realized gains, updates its own policy from realized outcomes, and does so without human intervention beyond strategic configuration. The construct composes three problems usually studied in isolation: sequential decision under uncertainty, online learning under non-stationarity, and capital allocation under irreversibility. The first has a clean theoretical ceiling (AIXI). The second is the subject of decades of online-learning literature. The third is the source of practical ruin for most agents that succeed in the first two.

The motivating scenario for IX is a recurring operator capital contribution of $5,000 per month, deployed by the agent under a 2.5% monthly target return (34.5% annualized), toward a terminal liquidity target of $10M after taxes. Under the stated parameters, the deterministic time-to-target is approximately 159 months (≈13.3 years) pre-tax and 173–184 months (≈14.4–15.3 years) after taxes depending on the realized short-term/long-term gains mix—a tractable horizon, and the 2.5%/month return assumption sits within the historically documented distribution of elite systematic performance, though sustained achievement across the full horizon is not commodity. Epistemic markers are used throughout: ✓ VERIFIED, ⊝ PLAUSIBLE, ◯ SPECULATIVE.

II. Theoretical Foundations

A. The AIXI Ceiling

AIXI defines the optimal action for an agent in any computable environment as the action that maximizes expected future reward, where the expectation is taken over all computable environment hypotheses weighted by their Kolmogorov complexity [1]. Concretely, the optimal action at time t is the argmax over a_t of the expectimax expansion across future observations and rewards, with the inner mixture over programs q running on a universal Turing machine U, each weighted by $2^{-\ell(q)}$ where $\ell(q)$ is program length [2]. ✓ AIXI is asymptotically optimal across the class of all computable environments. ✓ AIXI is uncomputable: Kolmogorov complexity is not a computable function.

B. The MC-AIXI-CTW Tractability Pattern

Veness, Ng, Hutter, Uther, and Silver [3] demonstrated that AIXI's two-slot structure survives two surgical substitutions. The world-model slot becomes Context Tree Weighting (CTW): a Bayesian mixture over variable-order Markov models bounded by depth D, which preserves the mixture-of-experts character of Solomonoff induction inside a tractable model class. The planner slot becomes Monte Carlo tree search (MCTS): expectimax is replaced by stochastic rollouts that concentrate compute on promising branches [4]. The composed system is computable on commodity hardware and learned non-trivial domains (Pac-Man, Kuhn poker, TicTacToe) from scratch.

C. The IX Substitution

IX preserves the two-slot pattern but specializes each slot for the RSF domain. The world model is a hybrid: a regime-aware state-space model over price and volume, an on-chain liquidity and orderflow model, and a finite Bayesian mixture over candidate trading hypotheses whose weights are updated by realized PnL—preserving the AIXI-style discipline of crediting theories that predict correctly and shrinking those that do not. The planner is MCTS guided by a policy network trained

TABLE I
IX SEVEN-LAYER ARCHITECTURE

Phase	Layer	Concern
0	Substrate	Data ingestion; on/off-chain unified feed; latency-aware reconciliation
1	World Model	Bayesian posterior over market regimes and competing hypotheses
2	Strategy	Policy generation under complexity-penalty regularization
3	Execution	Routing with information-leakage cost internalized
4	Risk	Recursive Kelly damped by self-calibration error; ruin-bounded sizing
5	Compounding	Real-time tax-aware reinvestment scheduling
6	Self-Improvement	Meta-learning grounded in live out-of-sample PnL
7	Compliance & Tax	After-tax reward signal; 1099-DA/8949 audit-trail by construction

via PPO [5] or its successor, with rollouts that evaluate not only expected return but the realized empirical ruin probability under each candidate action. The objective is therefore constrained:

$$\pi^* = \arg\max_{\pi \in \Pi} \mathbb{E}_\pi[R] \quad \text{subject to} \quad \Pr(\text{ruin} \mid \pi) \leq \varepsilon. \tag{1}$$

$\ominus$ The constrained variant is essential when capital is irreversibly lost on failure; the unconstrained AIXI objective is silently incompatible with leveraged or concentrated betting.

III. Architecture: The Seven Layers

IX decomposes into seven layers, each addressing a single concern. Inter-layer interfaces are explicit data contracts so that any single layer can be reimplemented or replaced without disturbing the rest (Table I).

Two architectural choices distinguish IX from a generic deep-RL trader. First, the risk layer implements recursive Kelly [6]: position size is the Kelly fraction multiplied by a damping coefficient that is a function of the agent's own recent calibration error, measured as the empirical Brier score on its forward predictions—an overconfident agent shrinks its own bet size automatically. Second, the self-improvement layer treats meta-loss as live realized PnL on small bounded capital, not backtested PnL, grounding self-modification in irreversible feedback rather than self-deception about a model that overfit the test set.

IV. Benchmark Framework: IX-Bench v0.1

IX-Bench evaluates candidate agents across eight metric families. Each family resolves to a 0–100 sub-score; subscores compose into a single IX Capability Score (IX-CS) with weights configurable per operational phase. Family 8—Probabilistic Target Achievement—is the family most often omitted by performance frameworks and is non-optional here (Table II).

TABLE II
IX-BENCH v0.1 METRIC FAMILIES

#	Family	Representative Metrics
1	Return	CAGR, TWR, IRR
2	Risk	MaxDD, DD Duration, VaR/CVaR, Risk-of-Ruin, Sharpe, Sortino, Calmar
3	Strategy Quality	Win Rate, Avg Win/Loss, Profit Factor, Recovery Factor, Tail Ratio, Info Ratio vs BTC-HODL
4	Agent Performance	Decision Latency, Brier Calibration, Tool Success, Hallucination Rate, OPE, Strategy Diversity
5	Capital Efficiency	Utilization, Reinvestment Velocity, Slippage, Fee Ratio
6	Recursion (RSF)	Gen-over-Gen Sharpe Δ, OOS Degradation, Walk-Forward, Regime Detection, Alpha Half-Life
7	Compliance/Ops	After-Tax Net, 1099-DA/8949 Accuracy [7], AML FP Rate, Audit Coverage, Custody Risk
8	Probabilistic Target	$\Pr$($10M after-tax by m175), $\Pr$(cumulative-return-negative at horizon), $\mathbb{E}$[time to target], return-distribution quantiles, monthly-return drawdown distribution, model misspec sensitivity

V. Implementation Pipeline

Each phase pairs a USPTO prior-art cluster—a region of patent space densely covered by existing claims—with a non-obvious gap and an open-source repository anchor that minimizes the rebuild surface (Table III). Build stack: Claude Code for primary implementation; Cursor for IDE iteration on Phases 1–2; Gemini for second-pass verification of Kelly derivations, ruin-probability arithmetic, and patent-claim novelty against the USPTO record; Claude (interactive thread) for architecture and document iteration.

VI. Risk, Reality, and Honest Limitations

Any whitepaper that omits the following is engaged in rhetorical performance rather than research.

A. The 2.5%-Monthly Calibration

✓ A 2.5% monthly compound return corresponds to a 34.5% annualized return. This sits below Renaissance Medallion's lifetime average of ∼40% annually net of fees but materially above commodity quant benchmarks: top-quartile systematic CTAs have historically realized 15–25% annualized; top-decile hedge funds 20–30%; the S&P 500 long-run ∼10%. $\ominus$ The 2.5%/month assumption therefore sits in the top decile of documented systematic performance over multi-year horizons—aggressive but not at the boundary of the historical record. $\ominus$ Achievability is more plausible at the $5,000/month scale than at fund scale: small-capital strategies can exploit microstructure regimes that do not scale, can deploy leverage flexibly without meaningful market impact, and face less alpha decay from public-record competition. ◯ Nonetheless, sustained 2.5%/month average across the full 159-month pre-tax horizon—through multiple regime transitions, asset rotations, and bear markets—is not commodity. Even Medallion has had down years. The architectural defenses against return-rate erosion are: regime-adaptive strategy switching (Phase 1's

TABLE III
IX IMPLEMENTATION PIPELINE: USPTO PRIOR-ART CLUSTERS AND OPEN-SOURCE ANCHORS

Phase	**USPTO Cluster (CPC)**	**Non-Obvious Gap**	**Open-Source Anchors**
0 – Substrate	G06Q40/04	Cross-feed disagreement treated as alpha signal, not noise to average away	ccxt, web3.py / ethers.js, clickhouse / arctic, polars / duckdb
1 – World Model	G06N5/04, G06N3/08	Explicit Bayesian posterior over competing market hypotheses; disagreement as exploration signal	hmmlearn, pytorch-forecasting, mlfinlab, vectorbt, tslearn
2 – Strategy	G06N3/126, G06Q40/00	Strategy synthesis under explicit Kolmogorov-complexity penalty as generalization regularizer	DEAP, PyGAD, stable-baselines3, Ray RLlib, FinRL
3 – Execution	G06Q40/04	Routing that internalizes the agent's own information-leakage cost; slippage treated as endogenous	hummingbot, jesse, freqtrade, 1inch SDK, cowswap
4 – Risk	G06Q40/06	Recursive Kelly damped by self-calibration error (Brier-score-driven sizing damping)	empyrical, pyfolio, riskfolio-lib, cvxpy, quantstats
5 – Compounding	G06Q40/06	Real-time pricing of marginal compounding opportunity vs. marginal tax cost	defi-sdk, aave-v3-sdk, backtesting.py
6 – Self-Improvement	G06N3/08, G06N20/00	Meta-loss grounded in live OOS PnL on bounded real capital, not backtest	Optuna, Ray Tune, NNI, MAML reference implementations
7 – Compliance	G06Q40/12	RL reward signal = after-tax net realized, not gross PnL (1099-DA-aware)	rotki, open IRS-form generators

mixture posterior); recursive Kelly with self-calibration damping (Phase 4) to compress drawdowns when overconfidence is detected; and tax-aware holding-period optimization (Phase 7) whose value grows in absolute terms as the return rate falls. The agent's responsibility under IX-Bench is to report the realized rolling return distribution honestly, not to claim deterministic achievement of the geometric mean.

B. Regulatory and Tax Surface

✓ Autonomous trading by U.S. persons is subject to CFTC, SEC, and FinCEN regimes depending on instrument class. ✓ Digital asset gains are reported on Form 8949 and Schedule D, with broker reporting under Form 1099-DA effective from 2026 [7]. ✓ Tax drag is material to the \$10M-after-tax target: assuming the top U.S. federal marginal bracket plus Alabama state income tax (5%), gains treated as long-term ($\geq$1 year holding) face an effective rate of $\sim$29% (20% federal LTCG + 3.8% NIIT + 5% state), while gains treated as short-term face $\sim$46% (37% federal ordinary + 3.8% NIIT + 5% state). The corresponding gross targets are \$14.1M and \$18.5M respectively, translating to 173 months at the long-term extreme and 184 months at the short-term extreme, against the 159-month pre-tax horizon. Active monthly rebalancing pushes most realized gains into the short-term regime by default; the Phase 7 compliance layer must therefore actively manage holding-period optimization to materially compress the time-to-target. Note that the tax-drag penalty in absolute months grows as the assumed return rate falls: an 11-month tax penalty at 2.5%/month corresponds to the same dollar tax burden as the 5–12-month penalty at 5%/month, but is more costly in calendar time because the underlying compounding is slower. ⊖ Operating an autonomous agent on behalf of any party other than the operator introduces fiduciary and registration questions whose resolution depends on jurisdiction and on whether the operator is acting as an investment adviser. The compliance layer (Phase 7) is non-negotiable and is therefore architecturally co-equal with the return-generating layers.

VII. CONCLUSION AND FUTURE WORK

IX is a research program, not a product. Its contribution is threefold: a clean theoretical lineage from AIXI through MC-AIXI-CTW to a domain-specific tractable agent; a benchmark framework that grades agents probabilistically and refuses to obscure the gap between target and feasible; and a seven-phase pipeline mapped to USPTO prior-art clusters and open-source anchors. Future work: derivation of the constrained-MCTS variant with formal ruin-probability bounds; empirical calibration of CTW depth for crypto microstructure; formal definition of the IX-RSI metric (Recursive Sharpe Improvement); and small-capital live trials with hard ruin-probability ceilings to populate Family 8 of IX-Bench with empirical posteriors.

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